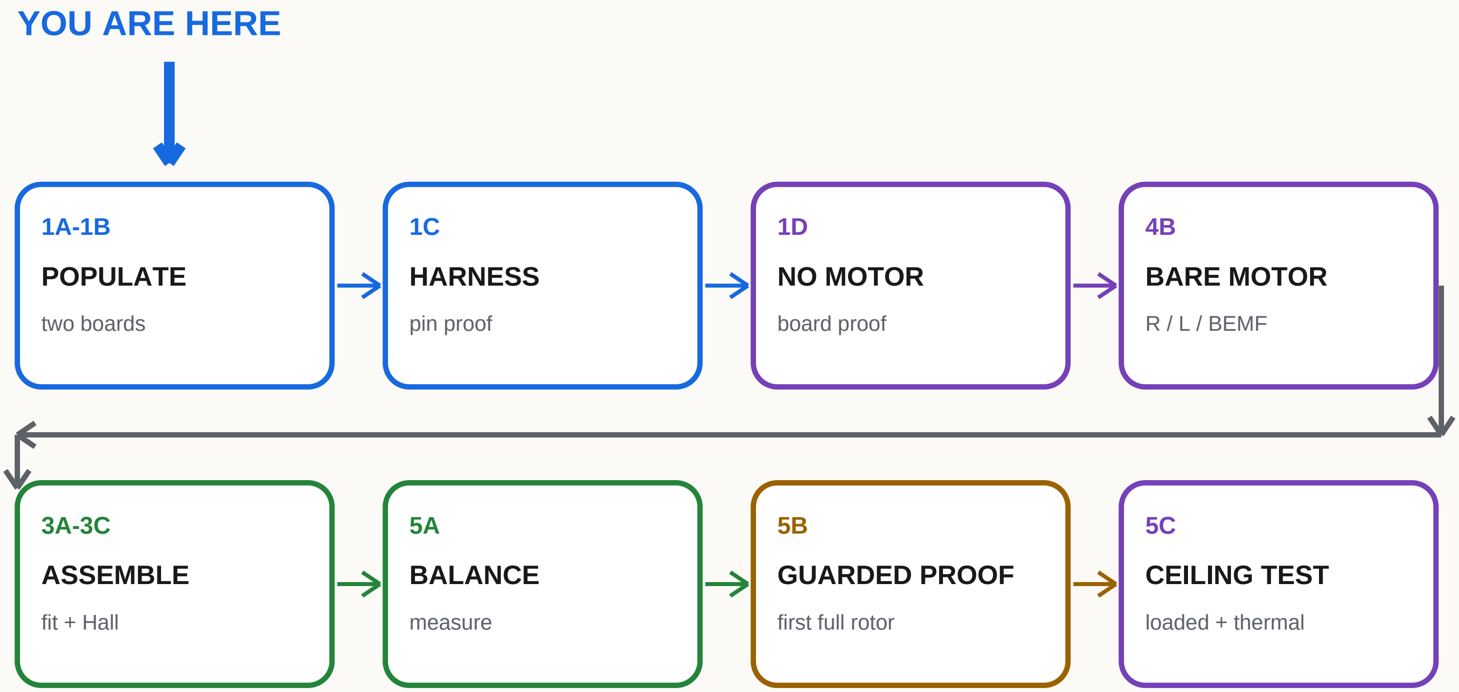
0A Active Integration Map

FIELD GUIDE



WHAT COUNTS AS DONE

- [ ] A result is measured and recorded, not just observed.
- [ ] Connector polarity and pin order are proved end-to-end.
- [ ] Balance + guarded proof precede ceiling loaded work.
- [ ] A failed gate stops the next powered stage.
- [ ] After loaded MPET, verify config before start/speed qualification.
- [ ] Normal safety firmware + long USB J6; cutoff outside sweep.

BOUNDARIES KEPT OUT OF THIS ACTIVE BINDER

Installed execution remains owner-managed. This binder records the agreed ceiling-commissioning sequence but does not prompt it. Deferred scope stays omitted unless explicitly reopened.

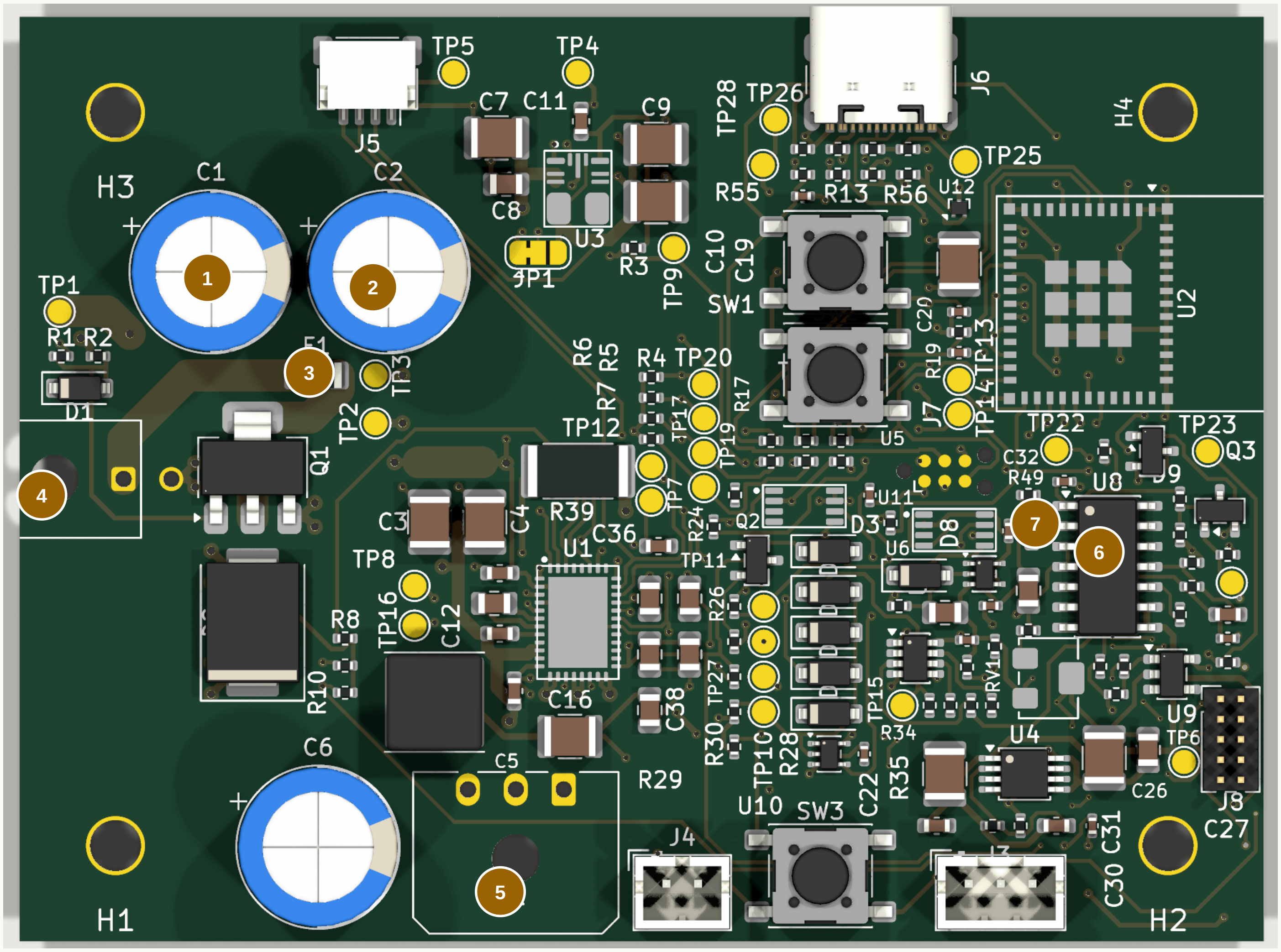
STOP / HANDOFF Start at 1A. Carry only the current sheet, board, and required tools to the bench.

SOURCE: docs/STATE.md; docs/integration.md; testing/test-matrix.csv

1A PCB-01 Population Map

FIELD GUIDE

TOP / COMPONENT SIDE - KiCad target appearance



1 C1	2 C2	3 F1	4 J1
5 J2	6 U8	7 C34	

SOLDER ORDER: U8 -> C34 -> J1/J2 -> C1/C2 -> INSPECT -> F1 LAST

1 + 2 C1 / C2

EEU-FR1H471, 470 uF 50 V. Positive lead to board +; negative stripe to pin 2 / PGND. Seat square without crushing bung.

3 F1 - INSTALL LAST

Low-profile tinned copper link across the 1206 pads. Near-zero ohms after soldering. External wall-side 3 A fuse remains mandatory.

4 + 5 CONNECTORS

J1 43045-0200; J2 43650-0300. Larger chisel tip. Tack one electrical pin, square and fully seat housing, then finish. Locating pegs are not soldered.

6 U8 / 7 C34

U8 LM2907M/NOPB: notch/dimple to pin-1 marker, tack pin 1 then diagonal pin 8, flux and drag-solder. C34 C1206C104K3GACTU: 100 nF C0G 25 V +/-10%, spanning the smaller 0603 site, not shorting it.

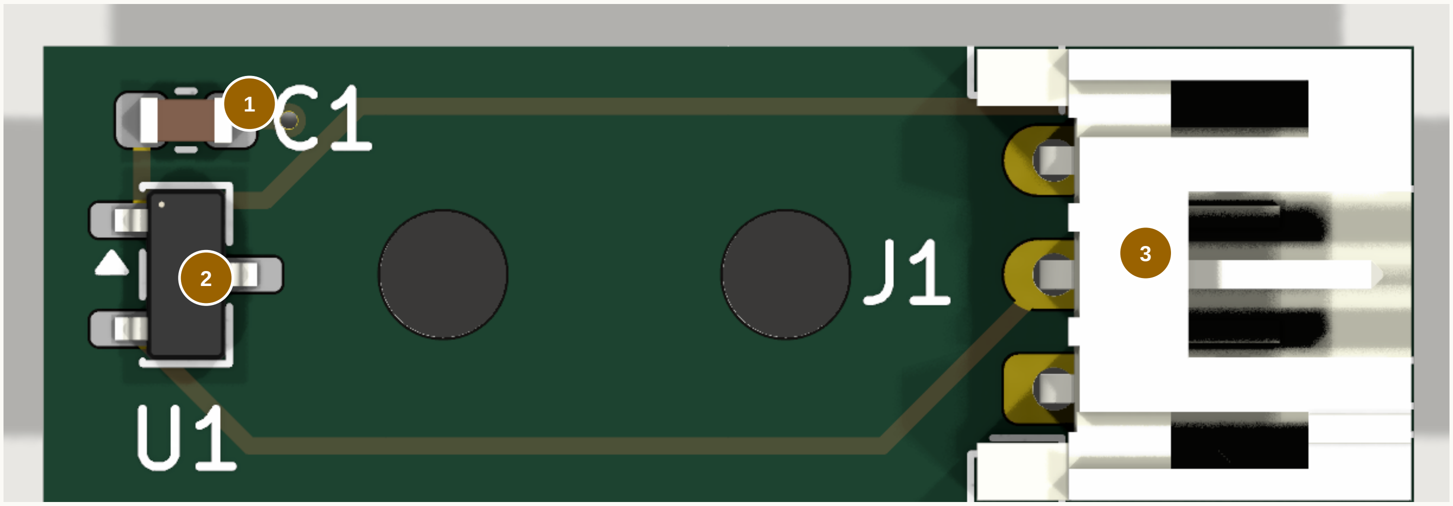
PCB-01: USE THE IRON, NOT BROAD HOT AIR. Then clean and dry; inspect under the microscope; verify no hard short from RAW24, VM24, 3V3, or 12V_TACH to ground; photograph the board.

SOURCE: pcb/pcb-01/pcb-01.kicad_pcb; bom/bom.csv; docs/electrical.md, SCH-06

1B PCB-02 Population + Soldering

FIELD GUIDE

TOP / MAGNET-FACING COMPONENT SIDE - KiCad target appearance



J1 / CABLE EXITS RIGHT →

1 C1
C0603C104K5RACTU, 100 nF 50 V X7R, 0603. Pin 1 / 3V3 is left; pin 2 / AGND is right.

2 U1
DRV5033FAQDBZR, SOT-23. Match package index to board triangle. Pin 1 upper-left = 3V3; pin 2 lower-left = HALL_TACH; pin 3 right = AGND.

3 J1
S3B-PH-K-S side-entry JST-PH. Top to bottom in this view: 3 AGND, 2 HALL_TACH, 1 3V3. Iron-solder last.

IRON METHOD: THIN SOLDER WIRE + FLUX

Flux + tin one pad → Place under microscope → Reheat to tack → Solder remaining pads → Inspect + wick bridges → Iron-solder J1

C1 - ONE-PAD TACK
Tin one pad lightly. Reheat it while placing C1, align the body, then solder the other end. Refresh the tack only if needed.

U1 - THREE LEADS
Flux, tack pin 3, correct alignment, then solder pins 1 and 2. Reflow the tack last if it needs a cleaner fillet.

J1 - LAST
Tack one pin, reheat while seating the body square and flush, finish the other pins, then revisit the tack.

FINAL INSPECTION

- ☐ All parts on magnet-facing side; sensor marked face will face magnet. Do not use magnet polarity as the orientation cue.
- ☐ Flux cleaned using its specified method; board completely dry.
- ☐ No short: 3V3 to HALL_TACH, 3V3 to AGND, or HALL_TACH to AGND.
- ☐ Microscope: wet fillets, no lifted lead, whisker, ball, or disturbed part; photograph board.

STOP / HANDOFF PCB-02 passes magnified inspection and unpowered three-net short checks. Do not power yet.

SOURCE: pcb/pcb-02/pcb-02.kicad_pcb; bom/bom.csv; docs/electrical.md, Hall daughterboard

1C Harness Build + Pin Proof

FIELD GUIDE

LOOK INTO THE MATING FACE; USE MOLDED PIN-1 MARKS, NOT WIRE COLOR MEMORY

POWER J1 - 18 AWG

RAW24	0V
1	2

Micro-Fit 2-way

DMM END-TO-END

MOTOR J2 - 18 AWG

W	V	U
1	2	3

Micro-Fit 3-way

HALL J3 / PCB-02 J1

3V3	TACH	AGND
1	2	3

PHR-3, straight-through

1-1 / 2-2 / 3-3

PROGRAM J7

3V3	TX	RX	EN	BOOT	GND
1	2	3	4	5	6

do not drive 3V3

EACH HARNESS: BUILD -> PROVE -> LABEL

Confirm housing view > Crimp + tug each contact > Insert to click > Continuity each pin > No cross-continuity > Label both ends

- [ ] Power: +24 V and 0 V preserved end-to-end.
- [ ] Hall: exactly 1-to-1, 2-to-2, 3-to-3.
- [ ] Motor: label positions W / V / U even if flying leads are not yet identified.
- [ ] Record length, gauge, wire colors, contact orientation, and result before sleeving.

J7 PIN 1 IS BOARD 3V3, NOT A POWER INPUT. Never let the programmer drive it while 24 V is connected.

STOP / HANDOFF Every harness is labeled, pull-tested, and signed off for pin continuity and no cross-continuity.

1D PCB-01 First Power, No Motor

FIELD GUIDE

WIRE THIS FIRST; PLACE PROBES BEFORE POWER

24 V BENCH SUPPLY
current limited

→

3 A FUSE + CUTOFF
within reach

→

PCB-01 J1
RAW24 / 0V

~~**J2 MOTOR**~~
~~DISCONNECTED~~

PROGRAMMER
TX / RX / EN / BOOT / GND; 3V3 crossed out

→

PROBES
DMM rails + scope VM with spring ground

NUMBERED TEST FLOW

1 RAILS
Brief power: 24 V, 3.3 V, AVDD, DVDD, DRVOFF; record current and heat.

2 HARDWARE CUTS
Force PGOOD, WDO, manual clear, OVERSPEED_N low one at a time -> DRVOFF rises without ESP help.

3 WATCHDOG
Static WDI: timeout 1.44-1.76 s, WDO pulse about 200 ms. 2 Hz falling edges prevent timeout.

4 STARTUP
Cold + slow-ramp power-up 20 times: U6 /PRE releases only after TACH_PGOOD_N.

5 RECOVERY
Remove/restore 24 V: disabled >=10 s; requires new command plus explicit arm.

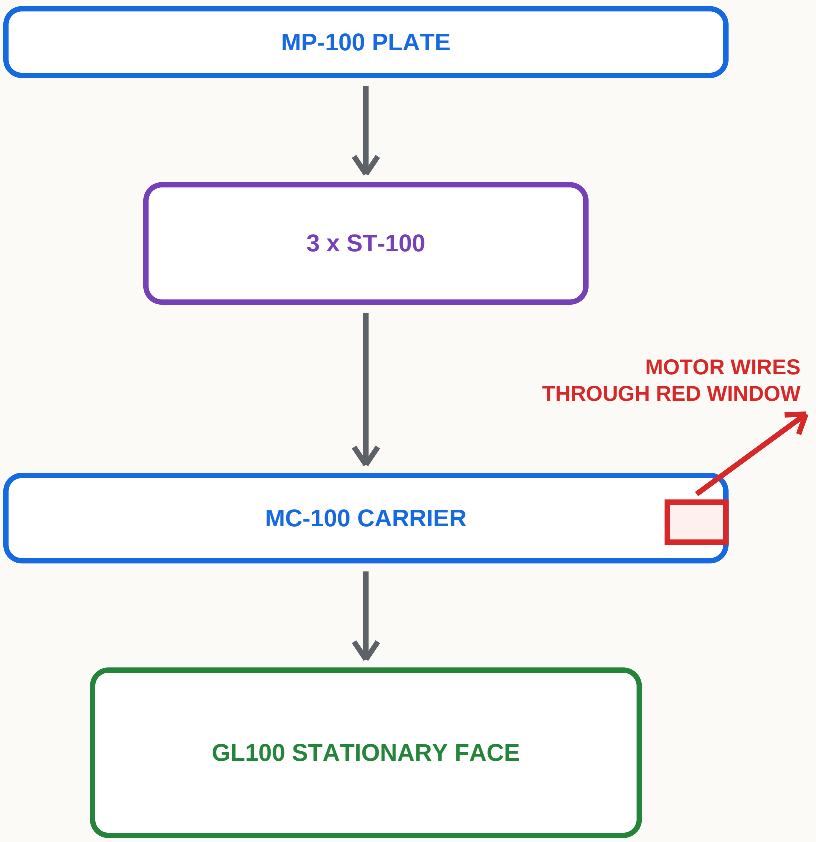
6 TACH
Bridge disabled. Inject 0-3.3 V, 1-3% duty, settle >=30 s: reset ~3.00 Hz, trip ~3.33 Hz.

PASS LIMITS: VM <=35 V target; 40 V rejects. Persistent tach lock clears only on a genuine low-voltage power cycle. Any automatic re-arm blocks release.

STOP / HANDOFF All listed board/safety tests have recorded passes. Do not connect the motor after any failure.

3A Stationary Stack Dry-Fit

EXPLODED SIDE VIEW - NON-POWERED



MP-100 -> ST-100
3 x M6 x 16 flat-head from plate ceiling face; heads flush or below.

ST-100 -> MC-100
3 x M6 x 20 A4-80 + Nord-Lock pairs. Hand-start, square seating.

MC-100 -> GL100
4 x M4 x 12 A4-80 into the 60 mm stationary pattern. The 50 mm face rotates.

THREAD DEPTH
Target about 5.5 mm engagement; 6.0 mm motor-thread maximum. A bottoming screw is felt at hand torque: stop and correct.

RELEASE CHECK

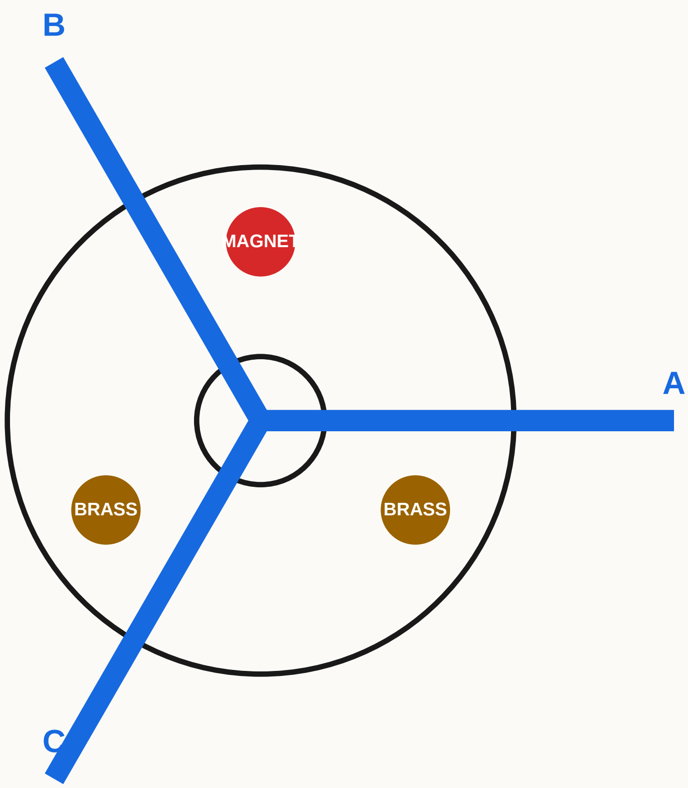
- ☐ All fasteners hand-start; no rocking or forced alignment.
- ☐ Owner-selected torque applied consistently; witness marks present.
- ☐ Stationary 60 mm and rotating 50 mm faces confirmed.
- ☐ Motor wire is unpinched and clear of sharp edges.
- ☐ Threadlocker only where specified; no screw bottoms.
- ☐ Stack remains non-powered until later gates pass.

STOP / HANDOFF Dry stack sits square without force, rocking, wire pinch, or bottoming screws.

SOURCE: docs/parts.md, GL100/ST-100/MC-100; bom/bom.csv

3B Rotor + Tach Inserts

FIELD GUIDE



HUB TO MOTOR

4 x M4 x 10 A4 flat-head from underside into 50 mm rotating pattern; heads 0.1-0.2 mm subflush. Pilot enters bore by hand only.

EACH BLADE

BP-100 v3 locating pins + captured M5 nuts; 4 x M5 x 22 per blade. Torque consistently and witness-mark.

THREE INSERT STATIONS

Epoxy retains one magnet + two brass slugs. Match complete insert-plus-epoxy station masses within 0.01 g after cure.

MASS + FINAL RECORD

MAGNET ______ g   **BRASS B ______ g**   **BRASS C ______ g**   **MAX SPREAD ______ g**

- ☐ Pilot hand-fit; all four hub heads subflush.
- ☐ Epoxy fully cured; inserts cannot move.
- ☐ Critical fasteners torqued and witness-marked.
- ☐ All three blade stations have four correct bolts and captured nuts.
- ☐ Hand-rotate several revolutions: no rub, click, or changing clearance.
- ☐ Photograph rotor before balance measurements.

Do not use cad/BP-100.step for current fit; it is stale v2 geometry.

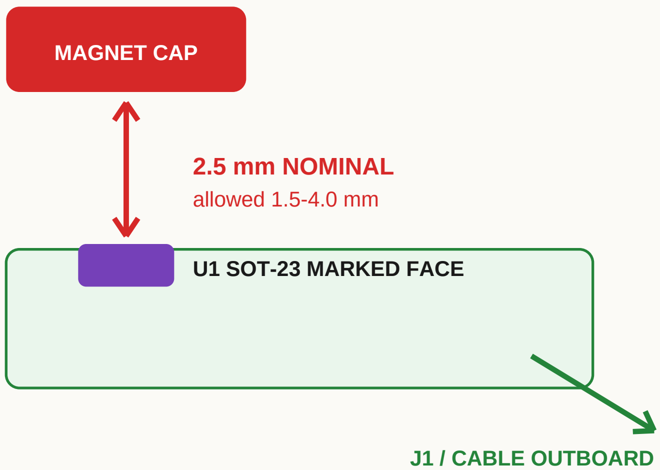
STOP / HANDOFF Complete rotor turns freely with correct retention; continue to 5A before powered motion.

SOURCE: docs/parts.md, RH-100/tach inserts; docs/blade-v2.md

3C Hall Gap + Electronics Bracket

FIELD GUIDE

SECTION THROUGH MAGNET AND SENSOR



COMMON CENTERLINE

Magnet and sensor element at radius 76.0 +/-0.5 mm.

PCB-02 MOUNTING

M2 holes on 6 mm pitch. At H1 use bare pan head or washer <=4.5 mm OD. Components face magnet.

BR-100

Strain-relieve Hall cable. Validate bracket around physical motor, board, magnet, and real gap.

PCB-01 BRACKET ENVELOPE

110 x 80 x 25 mm SERVICE ENVELOPE

Four 6-8 mm M3 standoffs. Keep isolated mounting holes clear of circuit ground. Independent PCB retention and cable clamps. Preserve connector and cable-bend keepouts.

HAND-ROTATION RELEASE

- ☐ One clean pulse/revolution with magnet.
- ☐ All cables and connector bodies clear rotor.
- ☐ No false pulse with magnet removed.
- ☐ Gap remains in range through full hand rotation.

STOP / HANDOFF Hall pulse is clean; retained PCB and clamped cables clear all moving parts.

4A Flash + One Persistent CLI Session

FIELD GUIDE

BUILD / FLASH / TALK

```
cargo build --manifest-path firmware/Cargo.toml
```

```
espflash flash --port /dev/cu.usbmodem2101 --non-interactive
  firmware/app/target/riscv32imac-unknown-none-elf/debug/stillair
```

```
firmware/target/debug/stillair --port /dev/cu.usbmodem2101 state
```

BARE DEV-BOARD INPUT JUMPERS

GPIO22 -> 3V3
PGOOD good

GPIO21 -> 3V3
nFAULT idle high

GPIO14 -> GND
ALARM idle low

SafeBoot → Starting → 15 s, no FG → NoRotation fault

EXPECTED ON A BARE BOARD; IT DOES NOT GENERATE FG EDGES

SESSION RULES

ONE READER
Stop raw-log capture before espflash or CLI opens the port.

USE SCRIPT
Multi-step simulator work must stay in one session; separate invocations reset it.

RELEASE GATE
config check must report config=verified. ok=true plus unverified is not a pass.

wait <state> --for <seconds> after asynchronous commands | stream <hz> --for <seconds> for CSV | config capture only after complete measured configuration

Never use raw reg write on a spinning motor except under an approved stopped/instrumented procedure. The simulator validates the harness, not the motor.

STOP / HANDOFF Firmware flashes; one persistent CLI session communicates and preserves scripted state.

SOURCE: AGENTS.md, Driving the fan; firmware/cli/src/main.rs

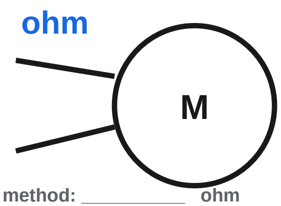
4B Desk: Bare GL100 Measurements

HOLD: IMAGE UNVERIFIED

NO BLADES x

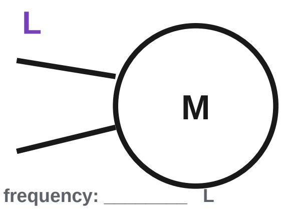
Guarded motor; manual cutoff ready; board/safety tests already passed

1 PHASE RESISTANCE



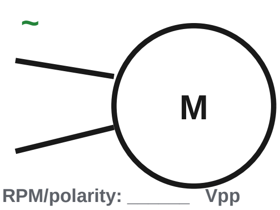
method: __________ ohm
U-V: ________________
V-W: ________________
W-U: ________________

2 PHASE INDUCTANCE



frequency: ________ L
U-V: ________________
V-W: ________________
W-U: ________________

3 MANUAL-SPIN BEMF



RPM/polarity: ______ Vpp
U-V: ________________
V-W: ________________
W-U: ________________

Repeat across phase pairs; record connection and polarity convention. Confirm 20 pole pairs.

GREEN LANE - DO NOW

- ☐ Measure R and L independently with method recorded.
- ☐ Only explicitly released limited rotation; keep current/speed caps.
- ☐ Manually spin only; scope line-to-line BEMF and convention.
- ☐ Keep loose hub hardware and blades off the motor.

AMBER LANE - HOLD

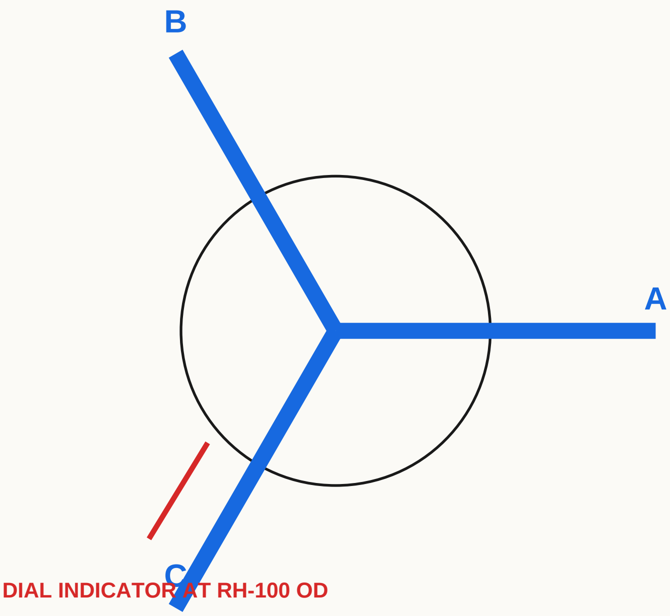
- ☐ No unloaded MPET; representative final rotor required.
- ☐ No config apply release until EEPROM completion/readback is proved.
- ☐ Golden image needs measured D-generation fields and review.
- ☐ Release requires config check: verified, not merely ok=true.

No live raw register writes. For any later powered work, VM <=35 V target; 40 V rejects.

STOP / HANDOFF R / L / BEMF records are complete. Loaded MPET and the golden image remain HOLD.

5A Rotor Balance + Runout Record

FIELD GUIDE



HUB RUNOUT
<=0.10 mm TIR

BLADE TIP HEIGHT
all three in one indexed setup; spread <=0.5 mm

FIRST MOMENT
each blade; spread <=0.5%

A / B / C MEASUREMENT TABLE

STATION	MASS (g)	FIRST MOMENT	TIP (mm)	CORRECTION
A				
B				
C				

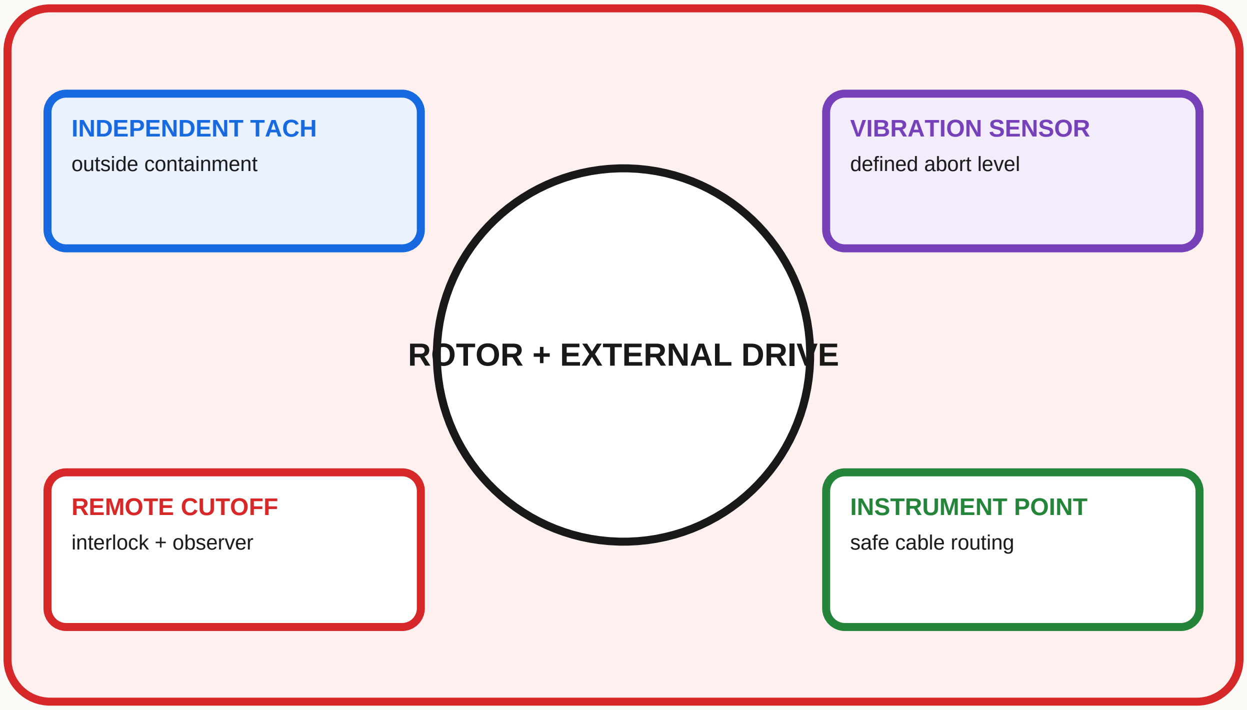
METHOD CHECK

- ☐ 0.01 mm indicator + 0.01 g calibrated scale.
- ☐ Correct only by documented slug method; remeasure.
- ☐ Define vibration rejection before powered motion.
- ☐ Hand-clearance pass and witness-mark photo saved.

STOP / HANDOFF MEC-05 passes numerically; do not advance a rotor that only looks balanced.

5B Guarded Rotor Proof

HOLD: PROCEDURE MISSING



OPERATOR A SAFE POSITION

OPERATOR B SAFE POSITION

216 RPM → **2 min CW** → **STOP + INSPECT** → **2 min CCW** → **STOP + INSPECT**

270 RPM = CALCULATION ONLY x NEVER DYNAMICALLY TEST

RELEASE BEFORE RUNNING

- ☐ Rated containment/fixture drawing released.
- ☐ Exact bypass state and restoration checks written.
- ☐ Pre-run balance, runout, clearance, witness marks recorded.
- ☐ Two-person roles, callouts, safe positions, duration defined.
- ☐ Independent RPM and measurable vibration/contact abort set.
- ☐ After bypass, reverify 180 RPM MCF and 200 RPM analog limits.

STOP / HANDOFF This sheet does not authorize the run. Release fixture and procedure first.

5C Ceiling Loaded Commissioning

HOLD: METHODS TO DEFINE

FINAL TEST CONNECTION

LAPTOP + CLI → LONG USB J6 → PCB-01 → MOTOR + ROTOR

NORMAL SAFETY FIRMWARE | 24 V VIA REACHABLE CUTOFF | CABLES OUTSIDE SWEEP

START MATRIX - PASSES / TARGET

DIRECTION	23.3 V (5)	24.0 V (20)	24.7 V (5)
CW			
CCW			

REJECT: retry, reverse kick, stall, hunting, or objectionable tonal sequence

LOADED MPET + VERIFIED CONFIG, THEN SPEED LADDER <=180 RPM

30 → 40 → 55 → 70 → 120 → 170 RPM

8-HOUR THERMAL RUN @ 170 RPM

	0h	1h	2h	3h	4h	5h	6h	7h	8h
AMBIENT C									
MOTOR C									
PCB C									
RMS A									
INPUT W									
ANOMALY									

CURRENT
<0.8 A normal | 1.0 A investigate | 1.5 A limiter not continuous

TEMPERATURE
motor <70 C | PCB <85 C

POWER
input <50 W; no dropout/overtemp

STOP / HANDOFF Starts, speed ladder, essential shutdowns, cooldown inspection, and thermal run all pass.